25901204 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECFS200 - VIRTUAL PRIVATE NETWORK SECURITY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Simplify the reasons for using VPN in companies. [An] / [CO1, CO2]
2. Define VPN protocols. [An] / [CO2]
3. Define 2FA . [E] / [CO1, CO4]
4. Explain AES-256 preferred in VPN. [An] / [CO2]
5. Explain about VPN server attacks. [An] / [CO1, CO2]
6. State any two features of TLs. [An] / [CO2]
7. Define RBAC. [A] / [CO3]
8. Describe the role of authentication in VPN. [An] / [CO2]
9. List the main steps in VPN connection setup. [U] / [CO1]
10. Identify the steps involved in 3DES encryption process. [A] / [CO2, CO3]
11. Define the key steps in user authentication. [E] / [CO1, CO4]

12. Explain Phase-1 negotiation in IPsec. [A] / [CO3]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

13. Differentiate between Man-in-the-Middle attack and eavesdropping attack. [U] / [CO2]

14. Explain the operations performed in IKEv2 and write the applications of IKEv2. [A] / [CO2, CO3]

15. Describe NAC in VPN environments. [A] / [CO2, CO3]

16. Write the steps to deploy remote access VPN. [A] / [CO3]

17. Compare Remote Access VPNs and SSL/TLS VPNs. [U] / [CO1]

18. Discuss about encryption algorithms used in VPNs. [A] / [CO2, CO3]

19. Explain the components of RADIUS architecture. [A] / [CO2, CO3]

20. Explain logging and monitoring techniques for VPN management. [A] / [CO3]

[5x6 = 30]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

21. Explain in detail the DES encryption process with all 16 rounds and diagram. [A] / [CO2, CO3]

22. Explain the integration of VPN with LDAP servers . [A] / [CO2, CO3]

23. Describe the types of VPN with examples. [U] / [CO1]

[10x2 = 20]